

CH5 Special Discrete Distributions

Bernoulli and Binomial R.V.

Bernoulli Definition

X is a Bernoulli trial (or Bernoulli random variable) with parameter p if sample space $S=\{s, f\}$, where s is called a success and f a failure and $X(s)=1$, $X(f)=0$.

The probability mass function of X is

$$p(x) = \begin{cases} p & x = 1 \\ 1 - p \equiv q & x = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = 0 \times P(X = 0) + 1 \times P(X = 1) = p$$

$$E(X^2) = 0^2 \times P(X = 0) + 1^2 \times P(X = 1) = p$$

$$Var(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p)$$

Binomial R.V. Definition

$$X \sim B(n, p)$$

X : The number of success

n : Perform n independently event

p : The probability of success

The p.m.f. of $X \sim B(n, p)$

$$p(x) = P(X = x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & \text{if } x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases}$$

$$E(X) = np$$

$$E(X^2) = n^2 p^2 - np^2 + np$$

If X is a binomial random variable with parameters $B(n, p)$, we can obtain

$$\sum_{x=0}^n p(x) = \sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x} = [p + (1 - p)]^n = 1^n = 1$$

Note: Multinomial theorem is used

Theorem

If X is a binomial random variable with parameter $B(n, p)$, then as k goes from 0 to n , $P(X=k)$ first increases monotonically and then decreases monotonically, reaching its maximum value when k is the largest integer less than or equal to $(n+1)p$.

假設我們進行 $n=100$ 次試驗，每次成功的機率 $p=0.665$ 。 $(n+1)p = 101 \times 0.665 = 67.165$

根據定義， k 是小於或等於 67.165 的最大整數，所以 $k=67$

Ex 5.2 A restaurant serves 8 entrees of fish, 12 of beef, and 10 of poultry.

If customers select from these entrees randomly, what is the probability that 2 of the next four customers order fish entrees?

$$X(4, \frac{8}{30})$$

$$P(X=2) = \binom{4}{2} \left(\frac{8}{30}\right)^2 \left(\frac{22}{30}\right)^{4-2}$$

Ex 5.2 The probability of a bit error in a communication line is 0.001. Find the probability that a block of 10000 bits contains two or more errors?

$$X(10000, 0.001)$$

$$1 - P(X=0) - P(X=1)$$

Ex 5.2 Consider a jury trial in which it takes 8 of the 12 jurors to convict; that is, in order for the defendant to be convicted, at least 8 of the jurors must vote him guilty. Assume that each of the jurors makes the right decision with probability p , while g represents the probability that the defendant is guilty. What is the probability that the jury makes a right decision?

情況 1: (有罪) $\rightarrow$ 至少 8 人做出正確判決

$$P(A|g) = P(A|g) \cdot P(g) = \sum_{k=8}^{12} C_k^{12} p^k (1-p)^{12-k} \cdot P(g)$$

情況 2: (無罪) $\rightarrow$ 至少 5 人判無罪

$$P(A|g^c) = P(A|g^c) \cdot P(g^c) = \sum_{k=5}^{12} C_k^{12} p^k (1-p)^{12-k} \cdot P(g^c)$$

Mean of Binomial random variable

If X is a binomial random variable

$$E(X) = np$$

$$E[X(X-1)] = n^2 p^2 - np^2$$

$$E(X^2) = n^2 p^2 - np^2 + np$$

$$Var(X) = E(X^2) - (E(X))^2 = np(1-p)$$

5.2 Poisson random variable

Definition

When $n \rightarrow \infty, p(x) \rightarrow 0$ in a binomial probability function, then $n \times p = \lambda$

A discrete random variable X with possible values $0, 1, 2, 3, \dots$ is called Poisson

with parameter λ , if $P(X = i) \rightarrow \frac{e^{-\lambda} \lambda^i}{i!}$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

$$E(X^2) = \lambda^2 + \lambda$$

This can approximate the situation that happen at the very low chance but large amount. If so the λ is the mean value of it.



Poisson random variable: Example

- **Ex 5.11** Suppose that, on average, in every three pages of a book there is one typographical error. If the number of typographical errors on a single page of the book is a Poisson random variable, what is the probability of at least one error on a specific page of the book?
- **Sol:** $E(X) = 1/3$ so $X \sim P(1/3)$
- $$P(X = n) = \frac{(1/3)^n e^{-1/3}}{n!}$$
$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-1/3} \approx 0.28$$

5.3 Other Discrete Random Variables

Geometric Random Variable

Do n times in an experiment while results is $\{\text{failed} \times (n-1), \text{success}\}$

$$X \sim \text{Geo}(p) = P(X = n) = (1-p)^{n-1} \times p, \quad n = 1, 2, 3$$

$$E(X) = \frac{1}{p}$$

$$E(X^2) = \frac{2-p}{p^2}$$

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Because the p.m.f of the Geometric Random Variable is like 等比數列 ($r=1-p$) 故得名

Exercies p33. TODO

Theorem (Memoryless Property)

$$P(X \geq n+m | X > m) = P(X \geq n)$$

Negative Binomial R.V.

Let X be the number of the experiments until the r^{th} success,
which contains k failure, the probability of success p is:

實驗次數: $n = k + r$

$$P(X = n) = C_{r-1}^{n-1} \cdot p^r \cdot (1-p)^k \quad \text{denoted: } X \sim NB(r, p)$$

$$E(X) = \frac{r}{p}$$

Hypergeometric R.V.

N : Total item count

D : Defectived item count

n : Drawn item count which less than D or $N-D$ $n \leq \min(D, N-D)$

X : be the number of defective items drawn

$$P(X = x) = \frac{C_x^D C_{n-x}^{N-D}}{C_n^N} \quad \text{for } x \text{ in } 0 \sim n$$

$$E(X) = n \cdot \frac{D}{N}$$

Which denoted: $X \sim \text{HGeo}(N, D, n)$

Summarize

Bernoulli R.V. (Bernoulli Trial)

實驗只有成功或失敗

$$P(X = k) = \begin{cases} p & k = 1 \\ 1 - p \equiv q & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = 0 \times P(X = 0) + 1 \times P(X = 1) = p$$

$$E(X^2) = 0^2 \times P(X = 0) + 1^2 \times P(X = 1) = p$$

$$Var(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p)$$

Binomial R.V. $X \sim B(n, p)$

實驗 N 次，有 r 次的成功的機率

$$P(X = x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & \text{if } x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases}$$

$$E(X) = np$$

$$E(X^2) = p^2 n^2 - p^2 n + pn$$

$$Var(X) = E(X^2) - (E(X))^2 = np(1 - p)$$

Negative Binomial R.V. $X \sim NB(r, p)$

前面都失敗，但是最後發生第 r 次成功

$$P(X = n) = C_{r-1}^{n-1} \cdot p^r \cdot (1 - p)^k$$

$$E(X) = \frac{r}{p}$$

Poisson R.V. $X \sim P(\lambda)$

估計極小機率在大量實驗的機率

$$P(X = i) \rightarrow \frac{e^{-\lambda} \lambda^i}{i!} \quad \lambda = np$$

$$E(X) = \lambda$$

$$E(X^2) = \lambda^2 + \lambda$$

$$Var(X) = \lambda$$

Geometric R.V. $X \sim Geo(p)$

前面都失敗，第 n 次才迎來首次成功

$$P(X = n) = (1 - p)^{n-1} \times p, \quad n = 1, 2, 3$$

$$E(X) = \frac{1}{p}$$

$$E(X^2) = \frac{2 - p}{p^2}$$

$$Var(X) = \frac{1 - p}{p^2}$$

Hypergeometric R.V. $X \sim HGeo(N, D, n)$

N 個物件標記 D 個，然後抽出 n 個物件有 x 個標記的機率

$$P(X = x) = \frac{C_x^D C_{n-x}^{N-D}}{C_n^N} \quad x \in [0, n]$$

$$E(X) = n \cdot \frac{D}{N}$$